Global Music Streaming Trends Analysis

Introduction :

As global demand for digital entertainment grows, music streaming platforms are racing to understand what drives listener engagement. The Global Music Streaming Listener Preferences dataset provides a comprehensive look into user preferences across regions, age groups, genres, devices, and listening behaviors. This data is crucial for product teams, content strategists, and marketing managers aiming to deliver personalized, platform-specific experiences. By analyzing these patterns, organizations can identify genre trends, optimize recommendation algorithms, and target specific audience segments with greater accuracy. Ultimately, the goal is to enhance user satisfaction, increase subscription rates, and improve content acquisition strategies in an increasingly competitive market.

Objective:

The objective of this analysis is to explore and visualize global listener preferences across music streaming platforms in order to uncover trends in genre popularity, platform engagement, user demographics, and device usage. By leveraging visualizations, this project aims to provide actionable insights that help streaming services personalize user experiences, improve content recommendations, and optimize marketing strategies for targeted audience segments.

Requirements :

To perform the Exploratory Data Analysis (EDA), the following requirements must be met:

Software and Libraries

- **Python (**version 3.8+**)**
- **Jupyter Notebook or any Python IDE** (e.g., VS Code)
- Required Python libraries:
    - **Pandas** for data manipulation
    - **Numpy** for numerical operationships
    - **Matplotlib** and **seaborn** for data visualization
- **Power BI** for advanced visualizations and dashboard creation

Prerequisites :

Before beginning this project, it is useful to have:

- **Python Fundamentals:** Understanding Python and data analysis libraries such as pandas, numpy, and matplotlib. You should be familiar with data structures such as dataframes and series.
- **Power BI basics:** Knowledge of Power BI tools such as chart creation, filtering, and interactive report development. You should be familiar with creating measures , columns and also writing DAX Functions.

What You'll Learn at the End of This Project

Upon finishing this project, you will:

- **Master Data Cleaning:** Learn how to deal with missing values, identify data kinds, and prepare datasets for analysis.
- **Perform exploratory data analysis (EDA).** Gain firsthand experience analysing data distributions, correlations, and significant insights from datasets.
- **Improve Data Visualisation Skills:** Create meaningful visualisations with Power BI and Python libraries.
- **Dax Function :** Create measures and write DAX functions in Power BI.
- **Reporting Skills:** Learn how to successfully communicate data insights via interactive dashboards.

Resources :

- **Global Music Streaming Trends and Listener Insights :** This dataset focuses on the user's demographics and preference in music. This dataset has been taken from an open source Kaggle.

Dataset Link

The dataset consists of 5,000 records and 12 attributes, capturing listener demographics, streaming behaviors, and preferences across different music platforms.

- **Power BI Documentation**:
  - Power BI Official Documentation
  - Power BI Tutorials

- **Python Documentation**:

- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation

Dataset Overview:

Key Attributes :

1. **User_ID** – A unique identifier assigned to each user. Each value represents a different user in the dataset.
2. **Age** – The age of the user, represented as an integer value.
3. **Country** – The country of residence of the user, stored as a categorical value (e.g., USA, Germany, Japan).
4. **Streaming Platform** – The music streaming service used by the user (e.g., Spotify, Apple Music, YouTube).
5. **Top Genre** – The most preferred music genre of the user (e.g., Pop, Rock, Jazz, Classical).
6. **Minutes Streamed Per Day** – The average number of minutes the user streams music daily, represented as an integer.
7. **Number of Songs Liked** – The total number of songs the user has liked on the platform, represented as an integer.
8. **Most Played Artist** – The artist whose songs the user listens to most frequently (e.g., Adele, Ed Sheeran, Post Malone).
9. **Subscription Type** – Whether the user has a free or premium subscription, stored as a categorical value (Free/Premium).
10. **Listening Time (Morning/Afternoon/Night)** – The time of day when the user listens to music most, categorized as Morning, Afternoon, or Night.
11. **Discover Weekly Engagement (%)** – The percentage of engagement with the platform's recommended playlist, represented as a floating-point percentage value.
12. **Repeat Song Rate (%)** – The percentage of times the user replays songs, represented as a floating-point percentage value.

How are these features useful ?

1. **User_ID :**
    a. Acts as a **unique identifier** for each listener.
    b. Helps in **tracking user behavior** over time.
    c. Useful for **personalized recommendations** and segmentation.

2. **Age :**
    a. Identifies **age-based music preferences**
    b. Helps in **targeted marketing** for different age groups.
    c. Useful for **subscription analysis** (e.g., Are younger users more likely to use free plans?).
3. **Country**
    a) Reveals **regional trends in music preferences**.
    b) Helps streaming platforms customize **country-specific playlists**.
    c) Identifies **market expansion opportunities** in different regions.

4. Streaming Platform

a. Identifies **which platform is most popular**.
b. Helps platforms understand **user loyalty and migration trends**.
c. Useful for **competitive analysis** between Spotify, Apple Music, YouTube, etc.

5. Top Genre

a. Shows **which music genres are most popular** among listeners.
b. Helps in **playlist recommendations** based on genre trends.
c. Useful for **predicting future trends in music**.

6. Minutes Streamed Per Day

a. Measures **user engagement** with the platform.
b. Helps determine **how often users stream music**.
c. Used for **identifying heavy vs. casual users**.

7. Number of Songs Liked

a. Shows **how active a user is** on the platform.
b. Helps platforms suggest **similar songs** based on likes.
c. Useful for **building personalized recommendation systems**.

8. Most Played Artist

a. Identifies **trending artists** among users.
b. Helps in **personalized playlist creation**.
c. Useful for **music industry insights** (e.g., which artists have the most engagement).

9. Subscription Type (Free/Premium)

a. Helps in **revenue analysis** (premium users generate more revenue).
b. Identifies **differences in engagement** between free and premium users.
c. Useful for **designing promotions and subscription upgrades**.

10. Listening Time (Morning/Afternoon/Night)

a. Determines **when users are most active** on the platform.
b. Helps in **ad placement** (e.g., running ads at peak times).
c. Useful for **creating time-specific playlists** (e.g., workout music in the morning).

11. Discover Weekly Engagement (%)

a. Measures **how well personalized recommendations are working**.
b. Helps in **improving AI-driven recommendation models**.
c. Useful for **enhancing user retention** through better content suggestions.

12. Repeat Song Rate (%)

a. Shows **which songs have high replay value**.
b. Helps in identifying **hit songs** that people listen to repeatedly.
c. Useful for **predicting song popularity** and ranking music trends.

Task to be performed :

1. Key Questions to Explore:

I. What are the most popular music streaming platforms?
II. How does age impact music preferences?
III. What are the most streamed genres and artists?
IV. How do free vs. premium users differ in streaming behavior?
V. What time of the day do users stream music the most?
VI. Are there any regional trends in music streaming preferences?